

EXPERIMENT 5a – DATA VISUALIZATION USING MATPLOTLIB

Aim

To create and analyze various data visualizations such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots using Matplotlib for effective data interpretation.

Procedure

Step 1: Import Libraries

Import Pandas and Matplotlib libraries required for data analysis and visualization.

Step 2: Load the Dataset

Read the tourism dataset using Pandas and store it in a DataFrame.

Step 3: Select Required Data

Select important columns such as Location, Visitors, Rating, and Revenue for visualization.

Step 4: Create Different Charts

Create Line Chart, Bar Chart, Pie Chart, Histogram, and Scatter Plot using the tourism data.

Step 5: Analyze the Results

Observe the charts and understand the distribution and relationship between tourism data.

Program

1. Import Libraries and Read Dataset

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Read dataset
```

```
df = pd.read_csv(r"C:\Users\kalap\Downloads\tourism_dataset (1).csv")
```

```
# Display first five records
```

```
display(df.head())
```

```
display(df.head())
```

	Location	Country	Category	Visitors	Rating	Revenue	Accommodation_Available
0	kuBZRkVsAR	India	Nature	948853	1.32	84388.38	Yes
1	aHKUXhjzTo	USA	Historical	813627	2.01	802625.60	No
2	dlrdYtJFTA	Brazil	Nature	508673	1.42	338777.11	Yes
3	DxmlzdGkHK	Brazil	Historical	623329	1.09	295183.60	Yes
4	WJCCQlepz	France	Cultural	124867	1.43	547893.24	No

2. Program 1: Line Chart – Tourist Visitors

Top 10 locations for better visualization

```
top10 = df.sort_values("Visitors", ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))
```

```
plt.plot(top10["Location"], top10["Visitors"], marker="o")
```

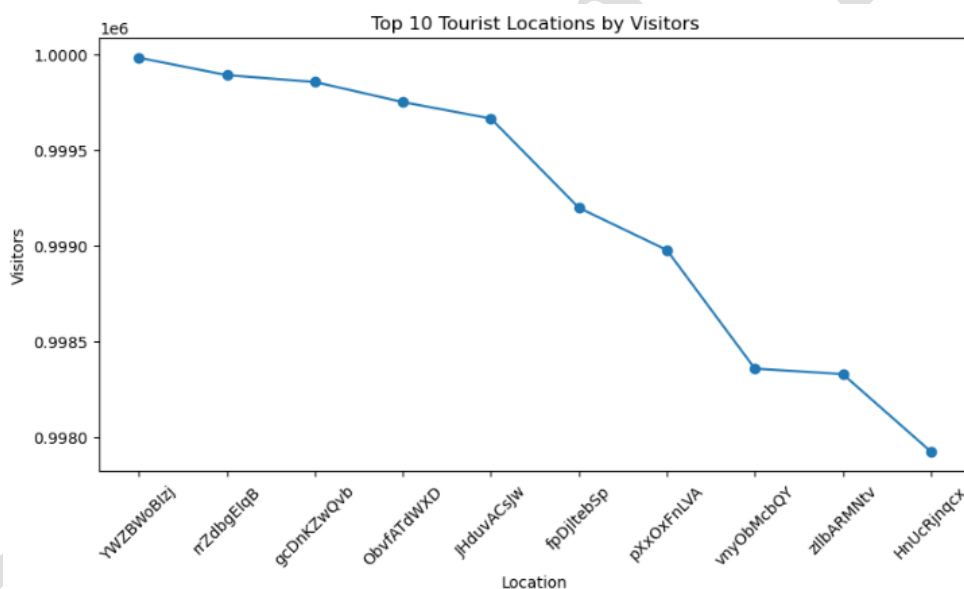
```
plt.title("Top 10 Tourist Locations by Visitors")
```

```
plt.xlabel("Location")
```

```
plt.ylabel("Visitors")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



3. Program 2: Bar Chart – Tourist Visitors

```
top10 = df.sort_values("Visitors", ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))
```

```
plt.bar(top10["Location"], top10["Visitors"])
```

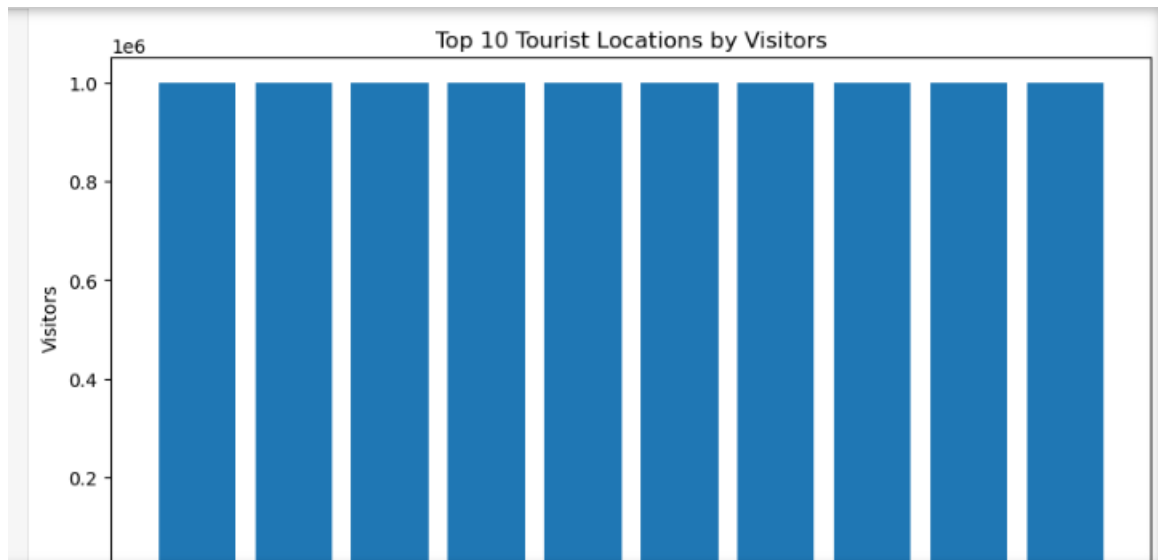
```
plt.title("Top 10 Tourist Locations by Visitors")
```

```
plt.xlabel("Location")
```

```
plt.ylabel("Visitors")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



4. Program 3: Pie Chart – Tourism Categories

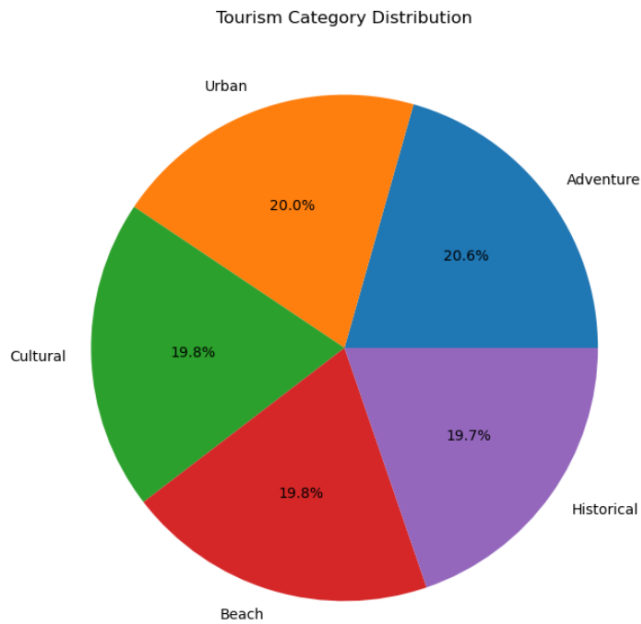
```
category_count = df["Category"].value_counts().head(5)
```

```
plt.figure(figsize=(8,8))
```

```
plt.pie(  
    category_count,  
    labels=category_count.index,  
    autopct="%1.1f%%"  
)
```

```
plt.title("Tourism Category Distribution")
```

```
plt.show()
```



8. Program 6: Multiple Line Chart – Visitors and Revenue

Select top 10 locations

```
top10 = df.sort_values("Visitors", ascending=False).head(10)
```

```
plt.figure(figsize=(10,5))
```

```
plt.plot(
    top10["Location"],
    top10["Visitors"],
    marker="o",
    label="Visitors"
)
```

```
plt.plot(
    top10["Location"],
    top10["Revenue"],
    marker="o",
    label="Revenue"
)
```

```
plt.title("Visitors and Revenue of Tourist Locations")
```

```
plt.xlabel("Location")
plt.ylabel("Value")
```

```
plt.xticks(rotation=45)
```

```
plt.legend()
```

```
plt.show()
```



RESULT: